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skill week-9

Task 1: To Implement how to List Active Jobs, Display Status Information, Retrieve Job IDs, Format Output, Update State Information, Test Job Listings

```
saichaitanya@GeethzzSai: ~/s × + v
saichaitanya@GeethzzSai:~$ mkdir skill_week9
saichaitanya@GeethzzSai:~$ cd skill_week9
saichaitanya@GeethzzSai:~/skill_week9$ nano 1.c
```

CODE:

```
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

#define MAX_JOBS 5

typedef struct {
    int job_id;
    pid_t pid;
    char state[20];
} Job;

Job jobs[MAX_JOBS];
int job_count = 0;

void add_job(pid_t pid) {
    jobs[job_count].job_id = job_count + 1;
    jobs[job_count].pid = pid;
    snprintf(jobs[job_count].state, 20, "Running");
    job_count++;
}

void update_jobs() {
    for (int i = 0; i < job_count; i++) {
        if (waitpid(jobs[i].pid, NULL, WNOHANG) > 0) {
            snprintf(jobs[i].state, 20, "Completed");
        }
    }
}

void list_jobs() {
    update_jobs();

    printf("\nActive Jobs:\n");
    printf("Job ID\tPID\tStatus\n");

    for (int i = 0; i < job_count; i++) {
```

```
GNU nano 8.7.1
}

int main() {
    pid_t pid1, pid2;

    pid1 = fork();

    if (pid1 == 0) {
        sleep(5);
        return 0;
    }

    add_job(pid1);

    pid2 = fork();

    if (pid2 == 0) {
        sleep(8);
        return 0;
    }

    add_job(pid2);

    printf("Two background jobs started.\n");

    list_jobs();

    sleep(6);

    printf("\nAfter updating job states:\n");
    list_jobs();

    waitpid(pid2, NULL, 0);

    return 0;
}
```

Output:

```
saichaitanya@GeethzzSai: ~/s  ×  +  v
saichaitanya@GeethzzSai:~$ mkdir skill_week9
saichaitanya@GeethzzSai:~$ cd skill_week9
saichaitanya@GeethzzSai:~/skill_week9$ nano 1.c
saichaitanya@GeethzzSai:~/skill_week9$ gcc 1.c -o 1
saichaitanya@GeethzzSai:~/skill_week9$ ./1
Two background jobs started.

Active Jobs:
Job ID  PID      Status
[1]     652     Running
[2]     653     Running

After updating job states:

Active Jobs:
Job ID  PID      Status
[2]     653     Running
saichaitanya@GeethzzSai:~/skill_week9$ |
```

Task 2: To design how to Identify Target Job, Transfer Terminal Control, Resume Processes, Wait for Completion, Update Job State, Test Foreground Switching.

```
saichaitanya@GeethzzSai: ~/s × + v
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
#include <signal.h>
#include <sys/wait.h>
#include <sys/types.h>

int main() {
    pid_t pid;
    int status;

    printf("Starting background job...\n");

    pid = fork();

    if (pid < 0) {
        perror("fork");
        return 1;
    }

    if (pid == 0) {
        setpgid(0, 0);

        printf("Child: Job started. PID = %d\n", getpid());

        raise(SIGSTOP);

        printf("Child: Job resumed in foreground.\n");

        sleep(3);

        printf("Child: Job completed.\n");

        return 0;
    }

    setpgid(pid, pid);
```

```
setpgid(pid, pid);

printf("Parent: Job PID = %d\n", pid);
printf("Parent: Job is stopped.\n");

waitpid(pid, &status, WUNTRACED);

if (WIFSTOPPED(status))
    printf("Parent: Job state = Stopped\n");

printf("\nSwitching job to foreground...\n");

kill(pid, SIGCONT);

printf("Parent: Job resumed.\n");

waitpid(pid, &status, 0);

if (WIFEXITED(status))
    printf("Parent: Job completed.\n");

return 0;
}
```

Output

```
saichaitanya@GeethzzSai:~/skill_week9$ nano 2.c
saichaitanya@GeethzzSai:~/skill_week9$ gcc 2.c -o 2
saichaitanya@GeethzzSai:~/skill_week9$ ./2
Starting background job...
Parent: Job PID = 731
Parent: Job is stopped.
Child: Job started. PID = 731
Parent: Job state = Stopped

Switching job to foreground...
Parent: Job resumed.
Child: Job resumed in foreground.
Child: Job completed.
Parent: Job completed.
saichaitanya@GeethzzSai:~/skill_week9$ |
```

Task 3: To Analyze Signal Delivery, Understand Signal Masks, Review Process Groups, Explore Terminal Signals, Test Signal Behavior.

```
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
#include <signal.h>

void handle_signal(int sig) {
    if (sig == SIGINT)
        printf("\nSIGINT received!\n");
    else if (sig == SIGTERM)
        printf("SIGTERM received!\n");
    else if (sig == SIGUSR1)
        printf("SIGUSR1 received!\n");
}

int main() {
    sigset_t set;

    signal(SIGINT, handle_signal);
    signal(SIGTERM, handle_signal);
    signal(SIGUSR1, handle_signal);

    printf("Process PID: %d\n", getpid());
    printf("Process Group ID: %d\n", getpgrp());

    printf("\nSignals are ready.\n");
    printf("Press Ctrl+C to test SIGINT.\n");

    printf("\nBlocking SIGUSR1 for 10 seconds...\n");

    sigemptyset(&set);
    sigaddset(&set, SIGUSR1);
    sigprocmask(SIG_BLOCK, &set, NULL);

    sleep(10);

    printf("SIGUSR1 is now unblocked.\n");

    sigprocmask(SIG_UNBLOCK, &set, NULL);
```

```

    printf("SIGUSR1 is now unblocked.\n");

    sigprocmask(SIG_UNBLOCK, &set, NULL);

    printf("Program running...\n");

    while (1)
        sleep(1);

    return 0;
}
|

```

OUTPUT:

```

saichaitanya@GeethzzSai:~/skill_week9$ nano 3.c
saichaitanya@GeethzzSai:~/skill_week9$ gcc 3.c -o 3
saichaitanya@GeethzzSai:~/skill_week9$ ./3
Process PID: 800
Process Group ID: 800

Signals are ready.
Press Ctrl+C to test SIGINT.

Blocking SIGUSR1 for 10 seconds...
SIGUSR1 is now unblocked.
Program running...
^C
SIGINT received!
^C
SIGINT received!
|

```

Task 4: To implement how to resume Stopped Jobs, Send Continue Signals, Update Job Status, Maintain Job Table, Verify Execution, Test Background Recovery.

```
saichaitanya@GeethzzSai: ~/s × + v
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
#include <signal.h>
#include <sys/wait.h>

typedef struct {
    int job_id;
    pid_t pid;
    char state[20];
} Job;

int main() {
    Job job;
    pid_t pid;
    int status;

    pid = fork();

    if (pid < 0) {
        perror("fork");
        return 1;
    }

    if (pid == 0) {
        printf("Child: Job started. PID = %d\n", getpid());

        raise(SIGSTOP);

        printf("Child: Job resumed.\n");

        sleep(3);

        printf("Child: Job completed.\n");

        return 0;
    }
}
```

OUTPUT:

```
saichaitanya@GeethzzSai:~/skill_week9$ nano 4.c
saichaitanya@GeethzzSai:~/skill_week9$ gcc 4.c -o 4
saichaitanya@GeethzzSai:~/skill_week9$ ./4
Parent: Job PID = 1071
Child: Job started. PID = 1071
Parent: Job state = Stopped

Switching job to foreground...
Parent: Job resumed.
Child: Job resumed.
Child: Job completed.
Parent: Job completed.
saichaitanya@GeethzzSai:~/skill_week9$ |
```